Braided Notions of Dialogue Categories: reworked proofs

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1 Dialogue Categories: Basic Constructions and Facts

Definition 1.1. Let $\mathcal{C}$ be a monoidal category, a $\mathcal{C}$ is *dialogue category* if there is an exponentiable object $\bot$, i.e., there are bijections natural in A and B

$$\varphi_{A,B} : \mathcal{C}(A \otimes B, \bot) \cong (B, A \multimap \bot)$$
$$\psi_{A,B} : \mathcal{C}(A \otimes B, \bot) \cong (A, \bot \circ\!\!- B)$$

Here are some useful combinators:

- The left evaluation map:

$$\mathrm{leval}_A = \varphi^{-1}_{A, A\multimap\bot} 1_{A\multimap\bot} : A \otimes (A \multimap \bot) \to \bot$$

- The right evaluation map:

$$\mathrm{reval}_A = \psi^{-1}_{\bot\circ\!\!- A, A} 1_{\bot\circ\!\!- A} : (\bot \circ\!\!- A) \otimes A \to \bot$$

- The composite left evaluation map:

$$\mathrm{lev}_{B,A} : B \otimes ((A \otimes B) \multimap \bot) \to (A \multimap \bot)$$
$$\mathrm{lev}_{B,A} = \varphi_{A, B\otimes((A\otimes B)\multimap\bot)}(\mathrm{leval}_{A\otimes B} \circ \alpha_{A,B,A\otimes B\multimap\bot})$$

- The composite right evaluation map:

$$\mathrm{rev}_{A,B} : (\bot \circ\!\!- (A \otimes B)) \otimes A \to (\bot \circ\!\!- B)$$
$$\mathrm{rev}_{A,B} = \psi_{(A\otimes B\circ\!\!-\bot)\otimes A, B}(\mathrm{reval}_{A\otimes B} \circ \alpha_{A\otimes B\circ\!\!-\bot, A, B})$$

Definition 1.2. Let $\mathcal{C}$ be a dialogue category,

- a *turn* in $\mathcal{C}$ is a natural isomorphism $\mathrm{turn}_A : A \multimap \bot \cong \bot \circ\!\!- A$.

*The original text is by Paul-André Melliès

- a *wheel* in $\mathcal{C}$ is a bijection $\mathcal{C}(A \otimes B, \bot) \simeq \mathcal{C}(B \otimes A, \bot)$ natural in both A and B.

Theorem 1.1. There is a 1-to-1 correspondence between turns and wheels in any dialogue category $\mathcal{C}$.

Proof. Given a turn $\mathrm{turn}_A : A \multimap \bot \cong \bot \circ\!\!- A$ natural in A, define $\mathrm{wheel}_{A,B} : \mathcal{C}(A \otimes B, \bot) \to \mathcal{C}(B \otimes A, \bot)$ as the following composite of bijections:

$$\mathcal{C}(A \otimes B, \bot) \xrightarrow{\varphi_{A,B}} \mathcal{C}(B, A \multimap \bot) \xrightarrow{\mathcal{C}(1_B, \mathrm{turn}_A)} \mathcal{C}(B, \bot \circ\!\!- A) \xrightarrow{\psi^{-1}_{A,B}} \mathcal{C}(B \otimes A, \bot)$$

This is obviously invertible, just take $\varphi^{-1}_{A,B} \circ \mathcal{C}(B, A \multimap \bot) \circ \psi_{A,B}$. wheel is also natural in both arguments. Take $f : A_1 \to A_2$ and $g : A_2 \otimes B \to \bot$, then the following square commutes:

$$\begin{array}{ccc} B \otimes A_1 & \xrightarrow{1_B \otimes f} & B \otimes A_2 \\ & {\scriptstyle \mathrm{wheel}(g \circ f \otimes 1_B)} \searrow \quad \swarrow {\scriptstyle \mathrm{wheel}_{A_2,B}(f)} & \\ & \bot & \end{array}$$

First of all, we have the following by the naturality of internal homs:

$$\mathrm{turn}_{A_1} \circ \mathcal{C}(f, \bot) = \mathcal{C}(f, \bot) \circ \mathrm{turn}_{A_2} \tag{1}$$

which following from the equality

$$\psi^{-1}_{B,A_1}(\mathrm{turn}_{A_1} \circ \varphi_{A_1,B}(g \circ f \otimes 1_B)) = \psi^{-1}_{B,A_2}(\mathrm{turn}_{A_2} \circ \varphi_{A_2,B}(g)) \circ (1_B \otimes f) \tag{2}$$

obtained by the following calculation:

$$\begin{aligned} \psi^{-1}_{B,A_1}(\mathrm{turn}_{A_1} \circ \varphi_{A_1,B}(g \circ f \otimes 1_B)) &= \psi^{-1}_{B,A_1}(\mathrm{turn}_{A_1} \circ \mathcal{C}(f, 1_\bot) \circ \varphi_{A_2,B}(g)) \\ &= \psi^{-1}_{B,A_1}(\mathcal{C}(f, 1_\bot) \circ \mathrm{turn}_{A_2} \circ \varphi_{A_2,B}(g)) \\ &= \psi^{-1}_{B,A_2}(\mathrm{turn}_{A_2} \circ \varphi_{A_2,B}(g)) \circ (1_B \otimes f) \end{aligned}$$

The rest of the proof of this clause is immediate.

The other way round part of the statement is more complicated. Assume we are given a bijection $\mathrm{wheel}_{A,B} : \mathcal{C}(A \otimes B, \bot) \to \mathcal{C}(B \otimes A, \bot)$ natural in both A and B. First of all, we have the following bijection:

$$\mathcal{C}(A \multimap \bot, A \multimap \bot) \to (A \multimap \bot, \bot \circ\!\!- A)$$

since

$$\mathcal{C}(A \multimap \bot, A \multimap \bot) \cong \mathcal{C}(A \otimes (A \multimap \bot), \bot) \cong \mathcal{C}((A \multimap \bot) \otimes A, \bot) \cong (A \multimap \bot, \bot \circ\!\!- A)$$

Similarly, there is a bijection $\mathcal{C}(\bot \circ\!\!- A, \bot \circ\!\!- A) \cong \mathcal{C}(\bot \circ\!\!- A, A \multimap \bot)$. $\square$